INCH-POUND
MIL-DTL-29405B(MC)
3 August 2021
SUPERSEDING
MIL-C-29405A(MC)
1 March 1985

DETAIL SPECIFICATION

CLOTH, BROADCLOTH: WOOL AND WOOL/NYLON

This specification is approved for use by the United States Marine Corps and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers wool and wool/nylon broadcloths used in dress uniforms (see 6.1).

1.2 Classification. The cloths are of the following types and classes (see 6.2).

1.2.1 Types.

a. Type I - Wool, 9.6 to 10.9 ounces per square yard.
b. Type II - Wool/nylon blend, 12.0 to 13.2 ounces per square yard.
c. Type III - Wool, 8.0 to 9.0 ounces per square yard.

1.2.2 Classes.

a. Class 1 - Scarlet 2501.
b. Class 2 - Dark Blue 2307.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-43665 - Cloth, Wool: Mothproofing Treatment of

Comments, suggestions, or questions on this document should be addressed to Marine Corps Systems Command, 2200 Lester Street, Quantico, VA 22134 ATTN: SEAL-SE-STDS or emailed to USMC_STDZ@usmc.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 8305

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DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-655 - Provisions for Evaluating Quality of Cloth, Wool, Worsted and Wool Blends

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL TRADE COMMISSION

The Wool Products Labeling Act of 1939

(Copies of this document are available online at https://ftc.gov.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC EP1	-	Evaluation Procedure for Gray Scale for Color Change
AATCC EP2	-	Evaluation Procedure for Gray Scale for Staining
AATCC EP8	-	Evaluation Procedure for AATCC 9-Step Chromatic Transference Scale
AATCC EP9	-	Evaluation Procedure for Visual Assessment of Color Difference of Textiles
AATCC TM8	-	Test Method for Colorfastness to Crocking: Crockmeter Method
AATCC TM15	-	Test Method for Colorfastness to Perspiration
AATCC TM16.3	-	Test Method for Colorfastness to Light: Xenon-Arc
AATCC TM20	-	Test Method for Fiber Analysis: Qualitative
AATCC TM20A	-	Test Method for Fiber Analysis: Quantitative
AATCC TM81	-	Test Method for pH of the Water-Extract from Wet Processed Textiles
AATCC TM132	-	Test Method for Colorfastness to Drycleaning
AATCC TM158	-	Test Method for Dimensional Changes on Drycleaning, Perchloroethylene: Machine

(Copies of these documents are available online at https://www.aatcc.org.)

ASTM INTERNATIONAL

ASTM D2130	-	Standard Test Method for Diameter of Wool and Other Animal Fibers by Microprojection
ASTM D3775	-	Standard Test Method for End (Warp) and Pick (Filling) Count of Woven Fabrics
ASTM D3776/D3776M	-	Standard Test Methods for Mass Per Unit Area (Weight) of Fabric
ASTM D5034	-	Standard Test Method for Breaking Strength and Elongation of Textile Fabrics (Grab Test)

(Copies of these documents are available online at www.astm.org.)

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CRC PRESS

Hayes, A. Wallace (editor) and Kruger, Claire L. (editor). *Hayes' Principles and Methods of Toxicology*. 2014.

(There are a number of ways through which this book may be obtained, such as online retailers, bookstores, public library, to name a few.)

OTHER PUBLICATIONS

Fabric Defect Replica Scales

(For information on obtaining copies of the Fabric Defect Replica Scales please contact U.S. Army Combat Capabilities Development Command (CCDC), Soldier Center, ATTN: FCDD-SCP-WC, 10 General Greene Avenue, Natick, MA 01760-5019.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Inspections.

3.1.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.

3.1.2 Conformance. When specified (see 6.2), a sample shall be subjected to a conformance inspection in accordance with 4.4.

3.2 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Standard sample. Unless otherwise specified in the contract (see 6.2), the finished cloth shall match the standard sample for shade and appearance (see 6.3).

3.4 Material. The material shall be a wool or wool blend, 2 x 1 right hand twill broadcloth.

3.4.1 Fiber.

3.4.1.1 Wool. The wool stock shall be of fleece or pulled wool, or a combination of both, no lower in grade than 60's U.S. Standard. The use of 30 percent noils is permitted. The use of reprocessed wools, reused wools, wool waste, or wool manufacturing byproducts other than those resulting from the supplier's own current production of this cloth is prohibited. Inspection shall be performed as specified in 4.5.1 and 4.5.2.

3.4.1.2 Nylon. The nylon fiber shall be a semi-dull, first quality nylon when tested as specified in 4.5.1.

3.4.2 Yarn. The wool (types I and III) and wool and nylon (type II) fibers shall be blended, carded, and evenly spun on the woolen system into singles yarn for both the warp and the filling when tested as specified in 4.5.1.

3.5 Color. The color shall be produced by stock or piece dying with dyes to match the standard sample for shade when tested as specified in 4.5.1 and 4.5.3.1. Light sources shall simulate artificial daylight under D_{75} illuminant with a color temperature of 7500±200 kelvin with illumination of 100±20 foot candles and under incandescent A illuminant with a color temperature of 2856±200 kelvin. The shade of the class 1 material shall be Scarlet 2501, and the shade of the class 2 shall be Dark Blue 2307. The shade shall be even and without variation when examined as specified in 4.5.3.3.

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3.6 Physical requirements. The finished cloth shall conform to the physical requirements listed in table I when tested in accordance with the test methods specified in 4.5.1.

TABLE I. Physical requirements.

Characteristic	Type I	Type II	Type III
Fiber content: body and selvage	95% wool (minimum)	75% wool (minimum), 25% nylon (maximum)	95% wool (minimum)
Weave	2 x 1 right hand twill	2 x 1 right hand twill	2 x 1 right hand twill
Weight, ounces per square yard	9.6–10.9	12.0–13.2	8.0–9.0
Yarns per inch (minimum)			
Warp	50	55	40
Filling	40	55	33
Breaking strength, pounds (minimum)			
Warp	50	83	40
Filling	40	80	35
Dimensional change to dry cleaning, percent (maximum)			
Warp	5.0	5.0	5.0
Filling	3.0	2.0	3.0
Colorfastness			
Light, after 40 AATCC Fading Units (AFU) or 170 kJ/(m^2nm) 1/ at 420 nm (minimum)	3–4	3–4	3–4
Perspiration (acid and alkaline) (minimum)			
Staining	3–4	3–4	3–4
Color change	3–4	3–4	3–4
Dry cleaning (minimum)			
Staining	3–4	3–4	3–4
Color change	3–4	3–4	3–4
Crocking (wet and dry) (minimum)	3.5	3.5	3.5
pH	4.0–8.0	4.0–8.0	4.0–8.0

FOOTNOTE:

1/ Kilojoules (kJ) per square meter per nanometer (m^2nm).

3.6.1 Selvage. To prevent the ends of the cloth from building up when rolled, the single yarn used in the selvage may be of a finer count than those used in the body of the cloth.

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3.7 Treatment. The fabric shall be thoroughly scoured, properly fulled, and free from plant matter (carbonized if necessary). The face shall be evenly sheared and pressed, and the appearance shall match the standard sample when tested as specified in 4.5.1 and 4.5.3.1.

3.7.1 Mothproofing. The cloth shall be given a mothproofing treatment in accordance with MIL-DTL-43665 when tested in accordance with 4.5.1.

3.8 Toxicity. When specified (see 6.2), the contractor shall furnish information that certifies that the finished product is composed of materials that have been safely used commercially or provide sufficient toxicity data to show compatibility with prolonged, direct skin contact when tested as specified in 4.6.

3.9 Roll identification. The back of the cloth shall be marked "BACK" at both ends of each roll with letters not less than ½ inch in height. The marking shall be clearly legible in any indelible marking used commercially and shall not show through on the face of the cloth when examined as specified in 4.5.3.4.

3.9.1 Fiber content marking. Each roll of finished cloth shall be labeled or ticketed for fiber content in accordance with The Wool Products Labeling Act of 1939 when examined as specified in 4.5.3.4.

3.10 Length, width, and put-up.

3.10.1 Length and put-up. Unless otherwise specified in the contract (see 6.2), the cloth shall be furnished in continuous lengths, each not less than 40 yards, and each length shall be put-up full width on a roll with the face side inward when examined as specified in 4.5.3.5.

3.10.2 Width. Unless otherwise specified in the contract (see 6.2), the width shall be 56 inches for types I and III and 60 inches for type II, excluding the selvage or tuck in or leno-type weave when woven on shuttleless looms.

3.11 Workmanship. The finished cloth shall be uniform in quality and shall conform to the quality of product established by this specification. The demerit points per 100 square yards, when calculated as specified in Section 4, shall not exceed the established maximum point value (see 6.2) and shall not adversely affect the serviceability, appearance, and uniformity of the product.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Conformance inspection (see 4.4).

4.2 Inspection conditions. Unless otherwise specified, excluded, amended, modified, or qualified in this specification or applicable procurement documents (see 6.2), all inspections shall be performed in accordance with all the requirements of referenced documents.

4.3 First article inspection. Unless otherwise specified in the contract (see 6.2), the first article inspection, submitted in accordance with 3.1.1, shall be examined in accordance with 4.5.1 through 4.5.3.3 for compliance with the design, configuration, workmanship, and testing as specified in this document.

4.3.1 First article samples and acceptance criteria. The sample size shall be as specified (see 6.2); the sample unit shall be 5 yards of cloth, full width; the lot size shall be expressed as yards of cloth; and the presence of excessive defects or failure of any testing shall be cause for rejection of the first article.

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4.4 Conformance inspection. Unless otherwise specified in the contract (see 6.2), conformance inspection, in accordance with 3.1.2, shall consist of the physical and chemical testing as specified in 4.5.1 and the visual examinations specified in 4.5.3.1 through 4.5.3.5.

4.4.1 Sampling and acceptance criteria for testing of physical and chemical characteristics. The sample size shall be in accordance with table II. The sample unit shall be 5 continuous yards, full width, of the finished cloth. The lot shall be unacceptable if one or more sample units fail to meet any requirement specified in4.5.1.

TABLE II. Sampling.

Lot size	Sample size (sample unit)
800 or less	2
801 through 22,000	3
22,001 and more	5

4.4.2 Sampling and acceptance criteria for visual inspection.

4.4.2.1 Shade and appearance inspection. Each roll in the lot shall be examined visually for shade and appearance (see 4.5.3.1). The sample unit shall be a 4-inch by 20-inch swatch of material, taken from each roll in the lot. A roll shall be unacceptable if it fails to meet the requirements for shade or appearance.

4.4.2.2 Yard-by-yard examination, shade variation, roll identification and marking, and length, width, and put-up. Sampling and acceptance criteria for yard-by-yard examination (see 4.5.3.2), shade variation (see 4.5.3.3), roll identification and marking (see 4.5.3.4), and length and put-up (see 4.5.3.5) shall be in accordance with MIL-STD-655. The fabric quality level for the cloth shall be as specified (see 6.2). In addition, the lot shall be unacceptable if two or more rolls in the sample are put-up with the face of the fabric facing outward.

4.5 Inspection methods.

4.5.1 Testing of physical and chemical characteristics. The cloth shall be tested for the characteristics listed in table III. The methods of testing shall be followed as specified in table III.

TABLE III. Physical and chemical characteristics verification.

Characteristic	Requirement	Verification
Wool grade	3.4.1.1	ASTM D2130 1/
Nylon fiber	3.4.1.2	AATCC TM20 2/
Yarn ply	3.4.2	Visual 3/
Shade	3.5	Visual 3/, 4/
Fiber content	3.6	AATCC TM20A 2/
Weave	3.6	Visual 3/
Weight	3.6	ASTM D3776/D3776M
Yarns per inch: warp and filling	3.6	ASTM D3775
Breaking strength: warp and filling	3.6	ASTM D5034
Dimensional change to dry cleaning: warp and filling	3.6	AATCC TM158

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TABLE III. Physical and chemical characteristics verification – Continued.

Characteristic	Requirement	Verification
Colorfastness		
Light after 40 AATCC Fading Units or 170 kJ/(m^2nm) at 420 nm	3.6	AATCC TM16.3, option 3 5/
Perspiration (acid and alkaline)	3.6	AATCC TM15 6/
Staining		
Color change		
Dry cleaning	3.6	AATCC TM132 6/
Staining		
Color change		
Crocking (wet and dry)	3.6	AATCC TM8 7/
pH	3.6	AATCC TM81
Appearance	3.7	Visual 3/, 4/
Mothproofing	3.7.1	8/
Toxicity	3.8	4.6

FOOTNOTES:

1/ Testing will be conducted on a scoured form. Failure of any specimen to meet the specified grade shall result in failure of the lot.

2/ See 6.6.

3/ One determination per sample unit and the results reported as “pass” or “fail”.

4/ AATCC EP9, option C (see 6.5), with sources simulating artificial daylight D_{75} illuminant with a color temperature of 7500±200 kelvin with illumination of 100±20 foot candles and under incandescent A illuminant with a color temperature of 2856±200 kelvin.

5/ Rated using the AATCC EP1.

6/ Rated using the AATCC EP1and AATCC EP2.

7/ Rated using the AATCC EP8.

8/ As specified in MIL-DTL-43665.

4.5.2 In-process inspection. Inspection shall be made at any point or during any phase of the manufacturing process to assure that only approved components are used (see 3.4.1). The Government reserves the right to exclude from consideration for acceptance any material for which in-process inspection had indicated nonconformance.

4.5.3 Visual inspections.

4.5.3.1 Shade and appearance examination on individual rolls. Each swatch taken from each roll in the lot shall be evaluated against the standard sample for shade and appearance in accordance with AATCC EP9, option C (see 3.5 and 6.5).

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4.5.3.2 Yard-by-yard examination and acceptance criteria. Each roll in the sample shall be examined yard-by-yard (full width excluding selvage) on the face side only with both overhead and back lighting. The defects are as defined in MIL-STD-655 and table IV, and only defects that are clearly noticeable at normal inspection distance (3 feet) shall be scored. Each defect shall be marked with a 1- to 1½-inch long string. The string shall be sewn into the selvage near the defect. A red string shall represent a three or four point defect, and a white string shall represent a one or two point defect. The defects are assigned demerit points in accordance with MIL-STD-655, with the exceptions listed below:

a. The defects found shall be counted regardless of their proximity to each other, except where two or more defects represent a single local condition of the cloth, in which case only the more serious defect shall be counted.

b. Only coarse yarns that exceed twice the normal diameter shall be scored.

c. Mixed filling shall be scored only if attributed in shade bar resulting from wrong ply, wrong twist in the yarn, or off shade yarn.

d. The defects specified in table IV shall be scored four points for each yard in which they occur.

TABLE IV. Four-point defects.

a. Any hole, cut, tear, or abrasion mark.
b. Edge ravels when pulled outward.
c. Uneven weaving throughout.
d. Broken or missing yarn.
e. Any spot, stain, or streak more than 1 inch in combined directions.
f. Knots greater than Fabric Defect Replica Scale level C (see 2.3).
g. Slubs greater than Fabric Defect Replica Scale level D (see 2.3).
h. Area of poor dye penetration, streak, or shade bar.
i. Width not as specified within a tolerance of no less than ½ inch.
j. Objectionable odor.
k. Slack or tight selvages.
l. Any embedded foreign matter or uncleanness.
m. Edges rolled, folded, scalloped, or corded, or baggy, ridgy, or wavy cloth.

4.5.3.3 Shade variation. During the yard-by-yard examination, each roll in the sample shall be examined for shade variation defects as specified in MIL-STD-655.

4.5.3.4 Roll identification and marking examination. During the yard-by-yard examination, each roll in the sample shall be examined for roll identification and marking defects as specified in MIL-STD-655.

4.5.3.5 Length and put-up examination. For Government procurements only, during the yard-by-yard examination, each roll in the sample shall be examined for length and put-up as specified in MIL-STD-655.

4.6 Toxicity test. When required (see 6.2), an acute dermal irritation study and a skin sensitization study shall be conducted. When the results of these studies indicate the material is not a sensitizer or irritant, a Repeat Insult Patch Test shall be performed in accordance with the Modified Draize Procedure (see Hayes' Principles and Methods of Toxicology). If toxicity requirement (see 3.8) can be demonstrated with historical use data on the finishing treatments used, toxicity testing may not be required (see 6.2). At a minimum, toxicity data should include results from a primary dermal irritation study in laboratory animals and a repeated insult human patch test (Modified Draize Procedure). The latter must be conducted under the supervision of a qualified dermatologist using at least 100 free-living individuals.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The cloths covered by this specification are intended for use in dress uniforms, insignia, stripes, and piping for dress uniforms worn by the United States Marine Corps.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. National Stock Numbers (NSNs).
c. Type and class required (see 1.2).
d. When first article is required (see 3.1.1).
e. When conformance inspection is required (see 3.1.2).
f. Standard sample (see 3.3).
g. Material toxicity information (see 3.8).
h. Length, width, and put-up required (see 3.10.1 and 3.10.2).
i. Maximum acceptable point values (see 3.11).
j. Inspection conditions (see 4.2).
k. First article inspection (see 4.3).
l. First article sampling and acceptance criteria (see 4.3.1).
m. Conformance inspection (see 4.4).
n. Fabric quality level (see 4.4.2.2).
o. When toxicity testing is required (see 4.6).
p. Packaging requirements (see 5.1).
q. A certificate of compliance, if acceptable for physical requirements (see 6.6).

6.3 Information requests. For access to information such as specifications, patterns, drawings, and standard samples of cloth, go to https://www.dla.mil/TroopSupport/ClothingandTextiles/SpecRequest.aspx for the Defense Logistics Agency Troop Support's Clothing and Textile Specification/Pattern/Drawing Request form. Complete the request form and then submit the form. Requests to use equivalent materials or components or to make changes to the pattern should be sent to the contracting officer for approval by the military service.

6.4 NSN for type 1, class 1. For informational purposes only, the NSN for type I, class 1 is 8305-01-209-6542.

6.5 Visual shade and appearance matching. In 2019, option A of AATCC E9, Visual Assessment of Color Difference of Textiles, was changed to option C. NOTE: The specimen plane and illumination source are parallel to each other and aligned so that the light flux is incident at the center of the specimen plane, which is set at a 35±5 degree angle relative to the horizontal. The observer will view the specimens at a 90 degree angle, relative to the plane of the specimens.

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6.6 Certificates of compliance. Certificates of compliance may be accepted for selected requirements (see 6.2). It is recommended that vendors verify acceptability prior to using certificates of compliance to address requirements.

6.7 Subject term (key word) listing.

Bloodstripe

Dress

Enlisted

Evening

Insignia

Marine Corps

Officer

Piping

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodian:
Navy – MC

Preparing activity:
Navy – MC
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Review activity:
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